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Balancing trust and distrust in generative AI chatbot adoption: a case study from China

平衡生成式 AI 聊天机器人采用中的信任与不信任：来自中国的案例研究

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ABSTRACT

Generative AI (GAI) chatbots are increasingly pivotal in transforming customer interactions across various industries, particularly in enhancing consumers' experience and operational efficiency in e-commerce. Drawing on dual factor theory, the study examines how consumer trust and distrust beliefs in GAI chatbots influence consumer satisfaction and continued use. The study's research framework is examined through collecting online data from Chinese consumers. The results indicate that perceived anthropomorphism, transparency, and efficiency positively influence consumers' trust beliefs. Meanwhile, privacy concerns and emotional detachment positively influence consumers' distrust beliefs. Consumer satisfaction and continued intention are positively influenced by trust and negatively influenced by distrust. Further, consumer satisfaction positively impacts continued intention. Finally, consumers' IT knowledge positively moderates the relationship between consumers' satisfaction and continued intention to use GAI chatbots. Several theoretical and practical implications reveal the need to make GAI chatbots more humanlike and responsive, as well as address consumers' concerns to establish trust.

摘要

生成式 AI (GAI) 聊天机器人在转变各行业客户交互关系中发挥关键作用, 尤其是在提升消费者体验和电子商务运营效率方面。基于双因素理论, 本研究探讨了消费者对 GAI 聊天机器人的信任与不信任信念对消费者满意度和持续使用意愿的影响。基于在线收集的中国消费者数据, 对本研究框架进行了检验。研究结果表明, 感知拟人化、透明性和效率正向影响消费者的信任信念。同时, 隐私顾虑和情感疏离正向影响消费者的不信任信念。消费者满意度和持续使用意愿受到信任的正向影响, 但受到不信任的负向影响。此外, 消费者满意度正向影响持续使用意愿。最后, 消费者的 IT 知识正向调节了消费者满意度与持续使用意愿之间的关系。本研究的理论和实践启示表明, 需要使 GAI 聊天机器人更加拟人化和反应迅速, 并解决消费者的顾虑, 从而建立消费者与 GAI 聊天机器人之间信任。

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1. Introduction

Technological advancements have dramatically reformed the working patterns of e-commerce platforms to offer better services to consumers (Shahzad et al., 2024b). For instance, traditional chatbots have long been used in customer support to provide automated responses and support. However, these chatbots are not always able to offer rich, humanlike interactions because they sometimes struggle to grasp the full context of a conversation (Chhabra et al., 2024; Luo et al., 2019). The advent of generative AI (GAI) chatbots has revolutionized customer support by providing 24/7, high-quality service. GAI chatbots leverage sophisticated language models and adaptive learning techniques to craft responses that are both personalized and context-aware, enhancing user engagement and overall satisfaction (Dwivedi et al., 2024).

GAI chatbots in e-commerce significantly enhance the shopping experience by assisting consumers in choosing products, placing orders, and resolving concerns (Chakraborty et al., 2024). In China, GAI chatbots are transforming customer interactions by driving innovation and improving user experiences across industries. For instance, Alibaba's GAI chatbot, Wenwen, generates personalized product recommendations and addresses customer inquiries with precision and speed (Utley, 2023). Similarly, Meituan's GAI chatbot, Wow, delivers tailored responses to customer questions, fostering convenience and trust (Post, 2023). Recent studies indicate that GAI chatbots are generally perceived as convenient and enjoyable, which encourages consumers to reuse GAI chatbots (e.g. Kim et al., 2024). GAI enhances relatability by making interactions more natural and offering customized replies that align with consumer profiles and preferences (Ameen et al., 2023; Christensen et al., 2024). Furthermore, GAI chatbots are competent in handling massive data quickly and accurately, resulting in a smooth customer experience (Dwivedi et al., 2024).

Despite these advancements, GAI chatbots are not without deficiencies. One major concern highlighted in prior studies is privacy, as these chatbots rely on access to consumers' personal information to interact, causing uncertainties regarding consent and data security (Mogaji & Jain, 2024; Ooi et al., 2024). Additionally, while GAI chatbots mimic human conversation, they lack genuine emotional understanding, leading to emotionally detached interactions (De Freitas et al., 2023). Consumers' emotional detachment can hinder the development of meaningful connections between consumers and the GAI chatbot, impacting their trust and satisfaction. Therefore, understanding the factors that shape trust and distrust in e-commerce using GAI chatbots is essential, as these beliefs directly impact users' willingness to continue engagement and determine the effectiveness of GAI chatbots. Distrust is a potentially different concept from trust and warrants investigation, considering its crucial and opposing impact (Moody et al., 2014). The dynamics of two factors that have contrasting impacts must be understood to avoid the confusion of cause and effect, which can be crucial to GAI chatbot success (Lee et al., 2015).

Prior studies have highlighted the importance of distinguishing between trust and distrust (Cho, 2006; Lee et al., 2015). Chakraborty et al. (2024) examined the role of trust in GAI chatbots, whereas distrust needs to be explored to understand GAI chatbot acceptance in e-commerce. Therefore, we intend to explore the impact of consumers' trust and distrust beliefs on their continued intention, with a particular focus on distinctive

GAI chatbot characteristics and consumers' concerns. In line with prior studies (Cenfetelli, 2004; Park & Ryoo, 2013; Talwar et al., 2021), we examined technology enablers and inhibitors using dual-factor theory (DFT). DFT is most appropriate for a comprehensive analysis, as it separately identifies and examines both positive (enabling) and negative (inhibiting) factors. It provides a balanced approach to decision-making by incorporating both trust-building and distrust-building behavior towards the GAI chatbot. Furthermore, Shahzad et al. (2024b) established the significance of information technology (IT) knowledge in enhancing the perceived value of technological advantages. Thus, consumers' IT knowledge may also influence their trust and distrust belief towards GAI chatbots. Consequently, this study answers the following questions.

RQ1: What are the key enabler and inhibitor factors influencing GAI chatbot adoption, and how do these factors impact consumers' trust and distrust beliefs?

RQ2: How does IT knowledge moderate the impact of trust, distrust, and consumer satisfaction on continued intention to use GAI chatbots?

By addressing these research questions, this study offers actionable insights for service providers and developers aiming to enhance GAI chatbot adoption. This study conducts qualitative analysis to determine key enabling (transparency, efficiency, personalization, and anthropomorphism) and inhibiting factors (privacy concerns and emotional detachment) that shape consumers' trust and distrust beliefs toward GAI chatbots. Building on these insights, this research employs a dual-factor approach to explore how enabling elements strengthen trust, while inhibiting factors lead to distrust. Additionally, the current study investigates the moderating role of IT knowledge, demonstrating how consumers with greater technological understanding interact differently with these factors. The findings present strategies for enhancing trust, reducing distrust, and improving chatbot interactions, ultimately fostering long-term consumer engagement.

The following sections will describe the literature review and outline the core principles underlying the model. We will present the results and the methodology in the following two sections. The last segment will discuss the key results, implications, and constraints of this study.

2. Literature review

2.1. Generative AI chatbots and e-commerce

GAI-driven chatbots are 'online services powered by AI algorithms that use large language models to help in real-time query resolution, content creation, marketing automation, and many other tasks' (Chakraborty et al., 2024, p. 3). GAI can play a pivotal role in transforming businesses by enhancing user experience and operational efficiency. The ability of GAI-based chatbots to provide highly personalized recommendations is one of their primary benefits (Christensen et al., 2024). In the e-commerce context, GAI chatbots can analyze consumers' past purchases, preferences, and broader market trends to recommend products and stores that align closely with their interests (Mogaji & Jain, 2024; Sadiq et al., 2024). In addition, GAI chatbots can provide clear, detailed information about products, including specifications, features, and consumer reviews, ensuring that consumers are able to make informed purchasing decisions. The GAI chatbot's high

level of responsiveness improves the overall consumer experience and builds trust since the GAI chatbot can provide reliable and comprehensive information.

GAI chatbots improve anthropomorphism, transparency, and personalization through continuous learning and adaptation. As these chatbots interact with more consumers, they gather valuable data that helps them refine their responses and predict consumers' needs more accurately (Chakraborty et al., 2024). Adaptive learning allows chatbots to become more effective over time, offering accurate recommendations and solutions. In this way, chatbots appear humanlike (anthropomorphism), transparent, efficient, and provide more relevant recommendations (personalization). Moreover, GAI can help e-commerce platforms stay relevant and appealing to their users by dynamically adjusting product recommendations and promotions based on current market trends (Prasad Agrawal, 2024; Zhou et al.). Additionally, GAI can forecast demand more accurately, enabling retailers to optimize inventory management and staffing by predicting peak shopping periods and popular products (Dwivedi et al., 2024). E-commerce can offer seamless, efficient, and personalized services by integrating GAI chatbot capabilities, meeting the high expectations of modern consumers, and setting new standards.

2.2. Dual factor theory (DFT)

DFT offers a valuable framework for understanding context-specific factors in the adoption and acceptance of innovations (Herzberg et al., 1959). Based on DFT, facilitators and hinderers are contrasting groups of factors that influence technology adoption, with each set contributing uniquely to user perceptions and behavior (Cenfetelli, 2004). Researchers widely accept DFT in different contexts to examine technology adoption and acceptance, such as mobile payment (Talwar et al., 2021), cloud computing (Park & Ryoo, 2013), and voice assistant (Malodia et al., 2023). In the context of GAI chatbots, DFT is particularly suitable because these systems evoke both positive and negative consumer perceptions. Enablers might include their core technological benefits, such as personalized recommendations, transparency, perceived anthropomorphism, and efficient customer support. The positive aspects of GAI chatbots encourage consumers to engage with the platform, fostering trust and satisfaction. Conversely, inhibitors, such as privacy concerns and emotional detachment, highlight the unique risks and challenges associated with AI-powered interactions in e-commerce. For instance, while GAI chatbots enhance efficiency and personalization, consumers may remain sceptical about how their sensitive data is handled or feel disconnected due to the chatbot's lack of empathy. These factors are independently impactful and must be considered simultaneously to understand their combined influence on user behavior (Cenfetelli, 2004).

Additionally, DFT incorporates the concept of negativity bias, which suggests adverse experiences tend to influence user attitudes more profoundly than favorable experiences (Cenfetelli & Schwarz, 2011). In the GAI chatbot context, misunderstandings in conversation, errors in recommendations, or perceived impersonality can overshadow the chatbot's benefits. Such negative experiences can overshadow the chatbot benefits, leading to distrust and potentially a negative assessment of the platform. Therefore, e-commerce businesses must prioritize addressing these inhibitors to mitigate their impact and maintain consumer trust. Finally, recent studies emphasize that examining both enablers and inhibitors together provides deeper insights into consumer attitudes toward innovations

like GAI chatbots (Malodia et al., 2023; Talwar et al., 2021). Therefore, DFT provides a comprehensive framework for predicting customer behavior toward GAI chatbots’ continued intention. Building on this foundation, the next section identifies key enablers and inhibitors through a qualitative study, further refining our understanding of their impact on consumer trust, distrust, satisfaction, and continued use.

3. Hypotheses formulation

The key variables influencing GAI chatbot adoption, including enablers and inhibitors, were identified using a qualitative approach. We collected 30 open-ended responses from WeChat and QQ, asking participants to answer the following question: ‘What are the key enabling/inhibiting factors that enhance GAI chatbot adoption? Please list 3–4 factors and briefly describe them.’ Thematic analysis was conducted following established methods (Kwok et al., 2022; Yan et al., 2023), involving detailed reviews to understand participants’ perspectives. To ensure reliability, two independent researchers coded significant statements and categorized ideas into enabler and inhibitor themes, with a third researcher resolving discrepancies. A binary coding system (‘Yes’ or ‘No’) was applied to determine the prevalence of subthemes, ensuring objectivity. The analysis revealed key enablers such as transparency, efficiency, personalization, and anthropomorphism, and inhibitors like privacy concerns and emotional detachment. These variables represent the most salient factors shaping consumer perceptions and experiences with GAI chatbots, which formed the foundation for this study’s framework.

Following DFT, the research framework categorizes these variables into enablers and inhibitors, which influence trust, distrust, and satisfaction, ultimately affecting the continued intention to use GAI chatbots (see Figure 1). Prior research also supports the inclusion of these enablers, such as personalization (Christensen et al., 2024; Dwivedi et al., 2024; Hyun Baek & Kim, 2023), perceived anthropomorphism (Chakraborty et al., 2024), Transparency (Hoff & Bashir, 2014; Seong & Bisantz, 2008; Shahzad et al., 2024a), and GAI chatbot efficiency (Dwivedi et al., 2024; Hyun Baek & Kim, 2023; Kim et al., 2024) in

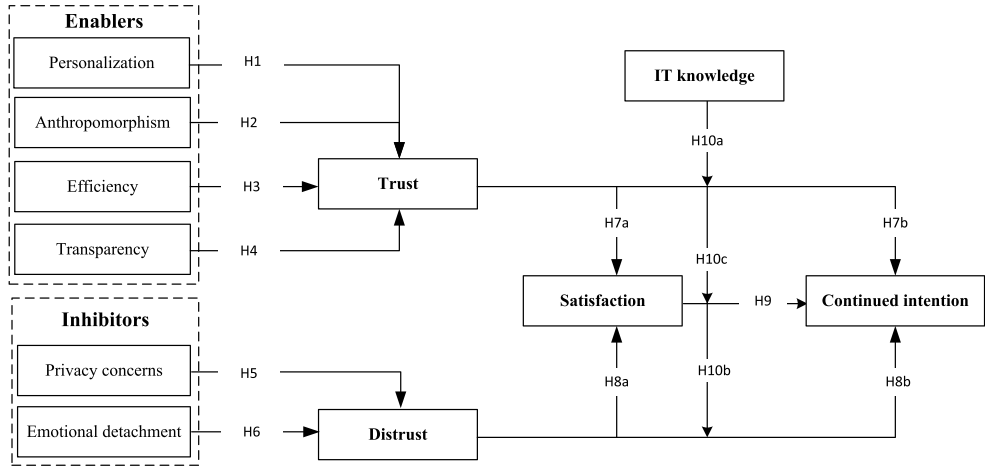


Figure 1. Research framework.

enhancing consumers' engagement. These enablers are well-established in prior literature as core facilitators of trust and user satisfaction in GAI-driven service contexts. Similarly, inhibitors such as privacy concerns (Mogaji & Jain, 2024; Ooi et al., 2024) and emotional detachment (De Freitas et al., 2023) may hinder consumer adoption of GAI chatbots. Recent studies also validate our inhibitor selection by identifying emotional discomfort and ethical concerns like privacy and transparency as key barriers to GAI adoption (Belanche et al., 2024; Vatankhah et al., 2024). In previous studies, enablers and inhibitors were examined on various contrasting factors, such as cost and benefits (Park & Ryoo, 2013) and negative and positive word-of-mouth (Talwar et al., 2021). Therefore, this study explores the impact of enablers and inhibitors on trust and distrust, providing deeper insights into consumer behavior. Numerous studies have also stressed the importance of using different measures to determine trust and distrust (Cho, 2006; Lee et al., 2015; Rufin et al., 2013).

Therefore, enablers are considered drivers of trust, while inhibitors are considered drivers of distrust. Based on DFT, trust and distrust themselves can also act as enablers and inhibitors in GAI chatbot adoption. Trust aligns with enablers by fostering positive perceptions and encouraging continued use, whereas distrust corresponds to inhibitors by amplifying concerns and discouraging adoption. These dynamics ultimately shape consumer outcomes, including satisfaction and continued intention to use GAI chatbots. Moreover, researchers established that consumers' IT knowledge substantially influences their perceptions of innovative technologies (Bassellier et al., 2003; Shahzad et al., 2024b). In the context of GAI chatbots, IT knowledge enables consumers to evaluate chatbot functionalities, interpret responses, and address concerns related to data privacy and algorithmic biases. Consumers with higher IT knowledge are better positioned to understand and assess the interplay of trust, distrust, satisfaction, and continued intention. Therefore, we include consumers' IT knowledge as moderating factors that may increase/decrease the impact of trust, distrust, and satisfaction on continued intention. Finally, we check the effect of demographic factors such as age, gender, education, and income. The following sections outline the theoretical rationale that underpins our hypotheses.

3.1. Personalization

Personalization refers to 'the ability to provide content and services that are tailored to individuals based on knowledge about their preferences and behaviors' (Adomavicius & Tuzhilin, 2005, p. 84). E-commerce platforms collect and analyse user browsing histories to make personalized offerings based on past transactions (Su et al., 2022). From the GAI chatbot's perspective, personalization encompasses how effectively the chatbot delivers tailored recommendations, responses, and interactions based on users' unique characteristics and past experiences. Personalized interactions create a sense of connection and rapport between users and the chatbot, enhancing the overall user experience and promoting long-term usage (Pillai et al., 2024).

Prior studies indicated that the perception of personalization is essential in shaping user trust and their willingness to adopt innovations (Komiak & Benbasat, 2006). Su et al. (2022) established that personalized interactions can make innovation more relevant and useful as they align closely with consumers' needs and preferences. Consumers tend to revisit an innovation that reliably fulfills their needs and expectations; thus, if they view

GAI chatbots as well-tailored, they are inclined to build trust beliefs aligned with their unique preferences. Building on this logic, we hypothesize that:

H1. Personalization positively influences trust beliefs.

3.2. Perceived anthropomorphism

Perceived anthropomorphism refers to ‘the assignment of humanlike attributes to non-human objects’ (Chan, 2020, p. 515). Perceived anthropomorphism can also be ‘considered a basic psychological process of inductive inference that can facilitate social human-nonhuman interactions’ (Blut et al., 2021, p. 634). In the context of GAI chatbots, perceived anthropomorphism involves features such as natural language processing, conversational style, and adaptive learning, which contribute to a humanlike experience.

Research suggests that higher perceived anthropomorphism fosters trust, as users tend to associate humanlike behavior with intelligence, reliability, and competence (Chakraborty et al., 2024). Studies have also shown that chatbots exhibiting anthropomorphic cues, such as personalized responses and emotive expressions, create a sense of social presence and emotional connection, which can enhance user engagement and reduce scepticism (Blut et al., 2021; Pizzi et al., 2023). However, excessive anthropomorphism may also create unrealistic expectations, leading to mismatches between user perceptions and actual chatbot capabilities (De Freitas et al., 2023). If a chatbot fails to deliver on its perceived humanlike qualities, it can result in disappointment, reduced trust, and increased user dissatisfaction (Pizzi et al., 2023). Therefore, maintaining an optimal balance between anthropomorphism and realistic chatbot performance is essential for maximizing trust. Thus, we hypothesize that:

H2. Perceived anthropomorphism positively influences trust beliefs.

3.3. Efficiency

Efficiency is defined as ‘how fast a service provider can fulfill customer needs, and focuses on the nature of service for the purpose of saving time’ (Xu et al., 2014, p. 421). In GAI chatbots’ context, efficiency refers to their ability to assist consumers in achieving their tasks quickly and accurately by delivering relevant information, answering queries, and facilitating transactions. Powered by machine learning algorithms and advanced natural language processing (De Freitas et al., 2023), GAI chatbots can efficiently navigate vast amounts of data to provide users with timely and contextually appropriate answers. For instance, when a customer seeks information about a product, a GAI chatbot can instantly retrieve and present detailed specifications, availability, and customer reviews, offering a faster and more convenient alternative to traditional customer service channels.

Research consistently highlights that efficient service delivery is a critical determinant of consumer trust (Hyun Baek & Kim, 2023). Efficient digital systems fulfill consumer needs seamlessly, instilling confidence in their reliability and accuracy (Lee et al., 2015; Xu et al., 2014). In the GAI chatbot context, timely and accurate query resolution enhances users’ perceptions of the chatbot as trustworthy and capable, strengthening their trust in the technology (Hsu & Lin, 2023). Therefore, we hypothesize that resolving consumer queries efficiently with GAI chatbots not only fulfills immediate needs but also

significantly boosts their trust, establishing them as a reliable and competent digital assistant in e-commerce.

H3. Efficiency positively influences trust beliefs.

3.4. Transparency

Transparency pertains to how clearly a system communicates its capabilities, limitations, and data-handling practices to users (Seong & Bisantz, 2008). For GAI chatbots, transparency includes explaining how data is used and providing clarity on the logic behind recommendations, which reduces uncertainty and builds trust. Prior studies emphasize the role of transparency in fostering consumers' trust (Hoff & Bashir, 2014; Seong & Bisantz, 2008). In e-commerce, Shahzad et al. (2024a) showed that platforms communicating clear data policies and recommendation logic tend to gain greater consumer trust. The transparency of GAI chatbots alleviates concerns about data misuse and the opaque nature of AI decisions, promoting perceptions of fairness and credibility. We therefore develop the following hypothesis:

H4. Transparency positively influences consumers' trust in GAI chatbots.

3.5. Privacy concerns

Privacy concerns refer to 'potential loss of control over personal information, such as when information about you is used without your knowledge or permission' (Featherman & Pavlou, 2003, p. 455). Regarding GAI chatbots, the importance of data collection and processing is heightened because these chatbots deliver personalized responses (Mogaji & Jain, 2024). GAI's advanced capabilities, such as profiling and behavioral prediction, require access to detailed user information (Ooi et al., 2024). Therefore, GAI chatbots can also increase privacy concerns, since chatbots need access to a wide range of personal data, such as shopping habits, preferences, and payment information. Prior studies have shown that heightened risk barriers and perceived risk can lead to increased consumer distrust (Malodia et al., 2023; Talwar et al., 2021). Thus, distrust may arise from the fear of potential misuse of personal information, which can overshadow the perceived benefits of using the service. Hence, we hypothesize that when consumers are anxious about how their private information is handled, they are more likely to view these chatbots with suspicion, doubting the platform's integrity and security.

H5. Privacy concerns positively influence distrust beliefs.

3.6. Emotional detachment

Emotional detachment refers to 'a strategy to deny the personal relevance of a potentially negative situation, which consequently makes the person feel psychologically detached from the situation and hence be emotionally less affected by the situation' (Chen et al., 2020, p. 279). In the context of GAI chatbots, emotional detachment reflects the lack of emotional connection or engagement between users and chatbots. This detachment arises when users perceive interactions as cold or impersonal, often due to the absence

of human warmth, empathy, or understanding (Laros & Steenkamp, 2005). While GAI chatbots excel at providing competent and logical responses, their inability to convey human emotions can leave users feeling disconnected and unengaged.

Research underscores the essential impact of emotional engagement in building trust and mitigating distrust. For example, Perrin-Martinénq (2004) found that emotional detachment substantially decreases consumer commitment and repurchase intentions, while Moon et al. (2017) reported that a lack of emotional engagement in digital interactions heightened consumer distrust. Therefore, we hypothesize that when consumers perceive a lack of empathy and emotional understanding in their interactions with GAI chatbots, they might question the chatbot's capability to fulfill their needs or resolve their concerns. This perception weakens trust in the chatbot's dependability, fostering distrust.

H6. Emotional detachment positively influences distrust beliefs.

3.7. Trust

Trust refers to 'the belief that the exchange party is able to fulfill its obligations reliably and confidently, is motivated to seek mutually beneficial gains, and will refrain from abusing the relationship' (Cho, 2006, p. 26). In the GAI chatbot context, consumers develop trust when they believe that the chatbots can provide accurate, reliable, and secure interactions that respect their privacy and personal data. Trust is a crucial factor in digital interactions, as it enhances perceived benefits and loyalty, encouraging users to engage with the technology (Chen et al., 2021; Prakash et al., 2023; Su et al., 2022). The advanced capabilities of GAI chatbots, such as their ability to understand natural language, personalize interactions, and handle complex requests, contribute to a positive user experience. GAI chatbots can enhance consumer trust by consistently delivering high-quality, accurate, and reliable responses to consumers' queries (Hyun Baek & Kim, 2023; Mogaji & Jain, 2024). Researchers have demonstrated that when consumers trust a digital platform, they become more satisfied with the service and develop continued usage intention (Hyun Baek & Kim, 2023; Klein & Martinez, 2023). Therefore, we anticipate that if consumers trust GAI chatbots, they are more likely to consider interactions as valuable and reliable, resulting in greater satisfaction and continued usage.

H7. Trust positively influences consumers' (a) satisfaction and (b) continued intention.

3.8. Distrust

Distrust refers to 'a belief that a partner will be incompetent, exhibit irresponsible behavior, violate obligations, and will not care about one's welfare or even intend to act harmfully' (Cho, 2006, p. 26). From the GAI chatbot's perspective, distrust can arise from several factors, such as perceived risks related to data privacy, the impersonal nature of interactions, or past negative experiences. Distrust is a key barrier to digital interactions, enhancing perceived risk and uncertainty and preventing consumers from using digital services (Chen et al., 2021; Lee et al., 2015). GAI chatbots can enhance consumer distrust in various ways. For instance, when chatbots provide inaccurate or irrelevant information, mishandle sensitive data, or fail to address user concerns adequately, users may doubt the

chatbot's reliability (Ooi et al., 2024). Furthermore, the lack of human empathy and personalized engagement can further disengage consumers, leading to feelings of detachment and distrust (De Freitas et al., 2023). Prior studies have shown that distrust can significantly reduce consumer loyalty and diminish their intention to continue using a service (Lee et al., 2015; Şengün & Wasti, 2011; Yang et al., 2023). Therefore, we propose that when consumers develop distrust towards GAI chatbots, their satisfaction is reduced, and their motivation to continue using these chatbots decreases.

H8. Distrust negatively influences consumers' (a) satisfaction and (b) continued intention.

3.8. Satisfaction

Satisfaction is 'the summary psychological state resulting when the emotion surrounding disconfirmed expectations is coupled with a consumer's prior feelings about the consumer experience' (Oliver, 2014, p. 6). It serves as a crucial factor in consumer behavior and loyalty, as customers who are satisfied tend to appreciate services and continue using them (Hsu & Lin, 2023; Shahzad et al., 2024c). Satisfaction reinforces positive attitudes by fulfilling expectations and enhancing overall user experience, making it a critical driver of continued intention (Shahzad et al., 2023a).

Established models and theories consistently highlight satisfaction as a primary predictor of continued usage. For example, the expectation-confirmation model posits that consumers experiencing satisfaction with a digital service tend to reuse it more frequently, as satisfaction lowers the perceived necessity of seeking alternative solutions and strengthens positive attitudes (Bhattacharjee, 2001). In the context of GAI chatbots, satisfaction arises from their ability to deliver convenience, efficiency, and personalization, which enhance perceived value and overall user experience. Therefore, we hypothesize that satisfied consumers tend to recognize and value the advantages provided by GAI chatbots, reinforcing their intention to continue using them.

H9. Satisfaction positively influences consumers' continued intention.

3.9. IT knowledge

Consumers' IT knowledge refers to 'their understanding of basic IT concepts and their awareness of IT in e-commerce' (Shahzad et al., 2024b, p. 4). IT knowledge enhances consumers' ability to evaluate and utilize emerging technologies, fostering more favorable attitudes toward innovation (Bassellier et al., 2003). Studies have shown that individuals possessing advanced IT knowledge tend to adopt and persist in using technology due to their ability to interpret, navigate, and leverage digital tools competently (Aggarwal et al., 2015; Devece, 2013). Conversely, consumers with limited IT knowledge often exhibit greater hesitation due to uncertainty or perceived complexity when interacting with AI-driven services. Recently, Shahzad et al. (2024b) demonstrated that greater IT knowledge improves consumers' ability to assess and extract value from technological innovations, such as blockchain applications.

In the context of GAI chatbots, IT knowledge can moderate the association between satisfaction and continued intention by influencing how consumers perceive and engage with chatbot capabilities. Consumers with greater IT expertise may find GAI

chatbots more efficient and beneficial, as they are better equipped to understand their functionalities, security mechanisms, and limitations. Their technological literacy allows them to navigate AI interactions more effectively, reducing frustration and enhancing overall satisfaction, which, in turn, reinforces their continued intention to use GAI chatbots. On the other hand, users with lower IT knowledge may struggle to interpret chatbot responses or trust the technology, leading to reduced satisfaction and a lower likelihood of continued use. Additionally, higher IT knowledge can mitigate distrust by alleviating concerns related to algorithmic biases, data privacy, and chatbot decision-making.

H10. IT knowledge moderates the relationships (a) between trust and continued intention, (b) between distrust and continued intention, and (c) between satisfaction and continued intention to use GAI chatbots.

4. Method

4.1. Survey formation

The survey instrument was developed based on validated scales from prior studies to ensure reliability and validity (see Appendix A). Each construct was carefully adapted to align with the unique context of GAI chatbots, ensuring that the measures accurately capture consumer perceptions and interactions. Modifications were made to improve clarity, relevance, and contextual fit, while maintaining the original theoretical integrity of the constructs. We measured personalization and transparency with four items each and were adopted from Lin et al. (2022) and Shahzad et al. (2024a). Anthropomorphism and efficiency were measured using five items each, modified from Balakrishnan et al. (2022) and Adam and Benlian (2024), respectively. In addition, four items of privacy concern and five items of emotional detachment were derived from Adam and Benlian (2024) and Perrin-Martineng (2004). Furthermore, trust and distrust were evaluated with three items each, adapted from Johnson (2007) and Johnson et al. (2008). IT knowledge was examined using five items, altered by Shahzad et al. (2024b). Finally, continued intention and satisfaction were assessed through seven items and modified from Bhattacharjee (2001). The constructs were measured on a 7-point Likert scale (7 'very strongly agree' to 1 'very strongly disagree'). After finalizing the questionnaire, we implemented a structured sampling approach to ensure responses from consumers who had actively engaged with GAI chatbots. The next section outlines the data collection process and sample demographics.

4.2. Data source

China holds the largest market share in the global e-commerce industry (Katharina, 2024). Prominent Chinese e-commerce companies such as Meituan, Alibaba, and JD.com readily embrace innovative technology to stay comparative and satisfy their consumers (Shahzad et al., 2023b; Shahzad et al., 2024b). Many Chinese e-commerce platforms have launched GAI chatbots to assist their customers. For instance, Meituan recently introduced a GAI chatbot named Wow that offers personalized responses to customer inquiries (Post, 2023). Moreover, e-commerce giant Taobao's consumers can also successfully use GAI

Table 1. Demographic characteristics.

Category	No (#)	%	Category	No (#)	%
Gender			Education		
Female	192	47.6	Diploma/other	134	33.3
Male	211	52.4	Bachelor	125	31.0
Age			Master	111	27.5
>25	123	30.5	PhD	33	8.2
25–35	107	26.6	Income (\$)		
36–45	98	24.3	Up to 1500	78	19.4
<45	75	18.6	1600–2500	96	23.8
			2600–3500	123	30.5
			<3500	106	26.3

chatbots to get assistance with products and services (Utley, 2023). Therefore, this study focused on Chinese consumers who had prior experience interacting with GAI chatbots. Due to language barriers, the questionnaire was translated into Chinese using expert review and back-translation, following Epstein et al. (2015) to ensure clarity and content validity. We collected data from prominent Chinese social media platforms, specifically QQ and WeChat. The survey consisted of three sections: the first served as a screening process to determine participant eligibility (i.e. the experience of using GAI chatbots), using nominal-scale (Yes/No) questions. Those who answered ‘yes’ continued with the survey, while those who answered ‘no’ were directed to the final page. The second section gathered demographic information, detailed in Table 1. The final section measured latent variables to validate the proposed hypothesis. Out of the 421 questionnaires collected, 18 were excluded during the screening due to incompleteness or consistent response patterns, resulting in a valid sample size of 403, which exceeds the required threshold ($403 > 138$), as per Tabachnick et al. (2007).

To ensure consistency and reliability, a pilot test was conducted with 50 respondents who had experience using GAI chatbots. Feedback indicated that the questionnaire was well-structured and easy to comprehend. The reliability of the questionnaire was assessed using Cronbach’s alpha (CA), with all CA values exceeding 0.8, thereby confirming the high reliability of the scale (Hair et al., 2010).

5. Results

5.1. Measurement model

The Amos and SPSS applications were used to analyse the data. Results of the measurement model indicate that all constructs are reliable and valid. The composite reliability (CR) and CA values for each construct are well above the recommended threshold of 0.8 (see Table 2), confirming the internal consistency of the survey items (Hair et al., 2010). Convergent validity is also established, as shown in Table 3, with average variance extracted (AVE) values exceeding the acceptable threshold of 0.5 for all constructs, indicating that the items adequately capture the underlying constructs (Hair et al., 2010). Table 3 illustrates that the Fornell-Larcker criterion confirms discriminant validity (Fornell & Larcker, 1981), which shows that the correlation between each construct’s indicators is higher than the correlation between other constructs’ indicators. Further, the fit indices for the measurement model, such as $CMIN/df = 1.566$, $CFI = 0.955$, $TLI = 0.950$,

Table 2. Validity and reliability.

Construct	Item	Loading	CA	CR	AVE
Personalization (PR)	PR-1	0.827	0.901	0.902	0.698
	PR-2	0.778			
	PR-3	0.865			
	PR-4	0.869			
Perceived anthropomorphism (PA)	PA-1	0.871	0.928	0.929	0.724
	PA-2	0.863			
	PA-3	0.851			
	PA-4	0.792			
	PA-5	0.875			
Efficiency (EF)	EF-1	0.858	0.905	0.905	0.656
	EF-2	0.833			
	EF-3	0.803			
	EF-4	0.762			
	EF-5	0.791			
Transparency (TP)	TP-1	0.764	0.835	0.838	0.564
	TP-2	0.794			
	TP-3	0.746			
	TP-4	0.698			
Privacy concerns (PC)	PC-1	0.841	0.899	0.899	0.690
	PC-2	0.846			
	PC-3	0.813			
	PC-4	0.822			
Emotional detachment (ED)	ED-1	0.808	0.912	0.913	0.676
	ED-2	0.844			
	ED-3	0.845			
	ED-4	0.823			
	ED-5	0.791			
Trust (TT)	TT-1	0.791	0.823	0.824	0.610
	TT-2	0.751			
	TT-3	0.800			
Distrust (DT)	DT-1	0.839	0.820	0.821	0.605
	DT-2	0.757			
	DT-3	0.733			
IT knowledge (ITN)	ITN-1	0.761	0.885	0.886	0.608
	ITN-2	0.739			
	ITN-3	0.831			
	ITN-4	0.789			
	ITN-5	0.776			
Satisfaction (SAF)	SAF-1	0.774	0.859	0.863	0.613
	SAF-2	0.850			
	SAF-3	0.764			
	SAF-4	0.739			
Continued intention (CI)	CI-1	0.831	0.899	0.903	0.756
	CI-2	0.872			
	CI-3	0.904			

RMSEA = 0.038, and SRMR = 0.036, are within acceptable limits, indicating a good fit (Shahzad et al., 2023b). These results support the measurement model's reliability and validity.

This study addressed concerns related to nonresponse bias, common method bias (CMB), and platform-specific differences between WeChat and QQ samples. To assess nonresponse bias, a comparison was made between the responses of the early and late participants, following the approach suggested by Armstrong and Overton (1977). At 5% significance, no differences were observed between the two groups. Additionally, a multigroup analysis was conducted to compare key constructs between the WeChat and QQ samples. The results showed no significant differences, confirming the

Table 3. Discriminant validity.

	PR	PA	EF	TP	PC	ED	TT	DT	ITN	SAF	CI
PR	0.836										
PA	0.022	0.851									
EF	0.194	0.290	0.810								
TP	0.045	0.053	0.029	751							
PC	-0.031	-0.115	-0.139	-0.028	0.831						
ED	-0.072	-0.097	-0.225	-0.009	0.450	0.822					
TT	0.057	0.311	0.353	0.253	-0.117	-0.112	0.781				
DT	-0.019	-0.148	-0.387	-0.148	0.356	0.377	-0.346	0.778			
ITN	0.020	0.075	0.028	0.039	-0.141	-0.174	0.107	-0.143	0.780		
SAF	0.002	0.103	0.111	0.444	-0.090	-0.057	0.396	-0.359	0.180	0.783	
CI	0.097	0.126	0.115	0.367	0.097	-0.047	0.471	-0.346	0.115	0.612	0.870

consistency of responses across platforms. For CMB, the study followed prior researchers and conducted Harman’s single-factor test (Chakraborty et al., 2024; Pillai et al., 2024). The results indicated that the first factor accounted for 17.074% of the variance, substantially lower than the 50% cut-off (Shahzad et al., 2023b), suggesting no CMB issue.

5.2. Hypothesis testing

Structural equation modelling (SEM) using AMOS was employed to evaluate the study’s hypotheses, including moderation effects analysed through interaction terms. SEM was chosen for several reasons. First, it is well-suited for confirmatory research, allowing for the validation of theoretically established relationships within a complex model (Sarstedt et al., 2021). Second, it accounts for measurement errors, enhancing the reliability and validity of latent constructs (Hair et al., 2010). Third, SEM enables the simultaneous estimation of multiple dependent relationships, offering a more comprehensive analysis compared to traditional regression methods that assess relationships individually (Hair et al., 2010; Sarstedt et al., 2021).

The results of SEM analysis indicate strong support for nine out of twelve hypotheses, as illustrated in Figure 2. Perceived anthropomorphism (H2, $\beta = 0.232$, $p = 0.000$), perceived efficiency (H3, $\beta = 0.320$, $p = 0.000$), and transparency (H4, $\beta = 0.272$, $p = 0.000$) positively influence consumers’ trust beliefs. Unfortunately, personalization is not significant (H1, $\beta = 0.022$, $p = 0.613$). In addition, privacy concerns (H5, $\beta = 0.256$, $p = 0.000$) and emotional detachment (H6, $\beta = 0.297$, $p = 0.000$) positively impact consumers’ distrust beliefs. As well, consumers’ trust beliefs have a positive effect on their satisfaction (H7a, $\beta = 0.357$, $p = 0.000$) and continued intention (H7b, $\beta = 0.238$, $p = 0.000$). In comparison, consumers’ distrust beliefs have a negative impact on their satisfaction (H8a, $\beta = -0.279$, $p = 0.000$) and continued intention (H8b, $\beta = -0.158$, $p = 0.001$). Additionally, consumers’ satisfaction positively impacts their continued intention (H9, $\beta = 0.475$, $p = 0.000$). Furthermore, consumers’ IT knowledge positively moderates the association between consumers’ satisfaction and continued intention (H10c, $\beta = 0.149$, $p = 0.000$). Moreover,

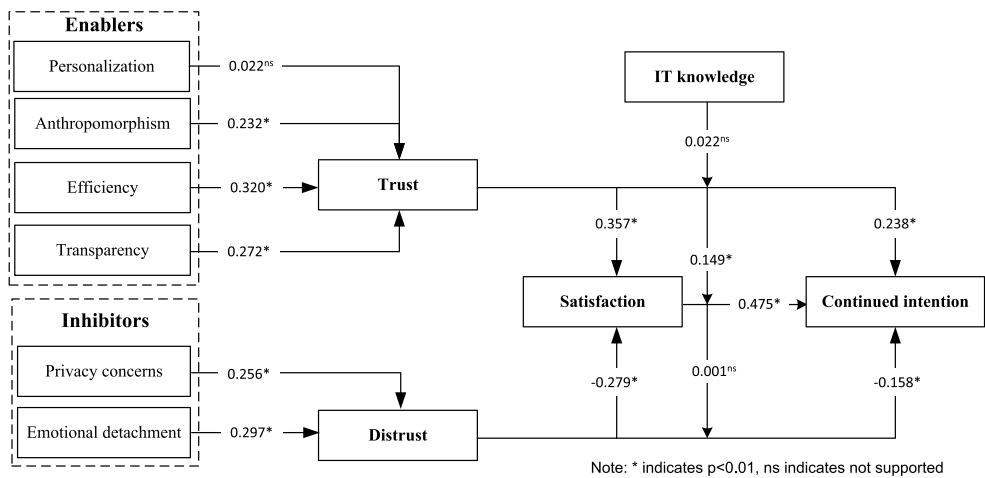


Figure 2. Structural model results.

H10a and H10b were not supported, and control variables did not have any impact on continued intention. Finally, continued intention explained 47.6% of the variance.

6. Discussion

Drawing on DFT, this study intends to advance insight into how enabler and inhibitor factors enhance consumers' satisfaction and continued intention to use GAI chatbots. The results established that perceived anthropomorphism, efficiency, and transparency positively influence consumers' trust beliefs. Respondents believe that GAI chatbots enhance their trustworthiness through their interactive capabilities, reliable service, and clear communication of data usage and decision-making processes. Transparency reduces uncertainty, as users who understand how GAI chatbots function are more likely to trust them (Hoff & Bashir, 2014; Seong & Bisantz, 2008). Additionally, GAI chatbots' capacity to deliver prompt and pertinent information strengthens consumer confidence (Chakraborty et al., 2024; Hsu & Lin, 2023). Furthermore, this study found insignificant results for personalization, despite prior research linking it with improved trust (Komiak & Benbasat, 2006; Su et al., 2022). A possible reason is that personalization has become a standard feature in digital platforms, making it less impactful on trust formation. In China's e-commerce landscape, where AI-driven personalization is already prevalent, consumers may not view it as a distinguishing factor.

Apart from enablers, the unique findings on inhibitors highlight the significant impact of privacy concerns and emotional detachment on consumer distrust. Consumers who highly value privacy tend to be more cautious and sceptical regarding the collection, storage, and utilization of their data (Featherman & Pavlou, 2003). This vigilance can amplify distrust, particularly if there is a perceived lack of transparency or control over personal information. Similarly, emotional detachment arises when GAI chatbots fail to exhibit warmth, making interactions feel impersonal or mechanical. As chatbots lack genuine empathy, users may perceive them as indifferent or incapable of understanding their needs, which further reinforces distrust (De Freitas et al., 2023). These findings suggest that both privacy concerns and emotional detachment function as inhibitors that weaken consumer confidence in chatbot interactions.

Furthermore, the results validate that consumers' trust beliefs positively impact their satisfaction and continued intention to use GAI chatbots. The results indicate that when consumers trust that the GAI chatbot will deliver accurate, reliable responses, they tend to experience greater satisfaction with their interactions. Trust enhances consumers' sense of security and confidence in chatbot interactions (Cho, 2006; Shahzad et al., 2024b), strengthening their commitment to using the technology consistently. Conversely, this study found that consumers' distrust beliefs negatively impact their satisfaction and continued intention to use. Distrust amplifies uncertainty and scepticism, as negative interactions diminish consumer confidence and deter continued chatbot use (Yang et al., 2023).

The results also established the positive influence of satisfaction on consumers' continued intention, aligned with prior studies (Bhattacharjee, 2001; Kim et al., 2024). Finally, this study establishes that IT knowledge moderates the relationship between consumer satisfaction and continued intention, while it does not significantly moderate the relationships between trust, distrust, and continued intention. The insignificance in the trust and

distrust relationships may stem from the fact that these constructs are primarily driven by intuitive perceptions and emotional reactions, which are less influenced by technical expertise. In contrast, IT knowledge significantly amplifies the influence of satisfaction on continued intention, as satisfaction reflects a positive evaluation of chatbot interactions that technologically experienced consumers are better equipped to assess. Their technical understanding allows them to derive significant value from chatbot functionalities and overlook minor shortcomings (Aggarwal et al., 2015; Shahzad et al., 2024b). Consequently, informed consumers are more likely to appreciate GAI chatbot benefits, enhancing their likelihood of continued use.

6.1. Theoretical implications

Researchers frequently investigate GAI across various domains due to its distinctive advantages. This study offers significant theoretical contributions by examining the potential of GAI chatbot adoption in e-commerce with a focus on how they can improve consumers' trust and distrust beliefs to enhance satisfaction and continued intention. First, the use of GAI chatbots in e-commerce is growing rapidly, and empirical studies are scarce. Numerous studies, such as Ooi et al. (2024) and Mogaji and Jain (2024), highlighted the importance of GAI and raised several questions to explore its significance in e-commerce. Thus, the first contribution is to respond to the research calls made by Ooi et al. (2024) and Mogaji and Jain (2024). Second, prior studies used different models and theories, such as the technology acceptance model (Kim et al., 2024) and the status quo bias theory (Chakraborty et al., 2024). However, the DFT was overlooked despite its ability to provide a comprehensive understanding of both enablers and inhibitors within a unified framework. Thus, the second contribution of this study is to explore GAI chatbot services in light of DFT.

Third, this study conducted a qualitative analysis to identify transparency, efficiency, personalization, and anthropomorphism as key enablers, and privacy concerns and emotional detachment as key inhibitors, shaping consumers' trust and distrust beliefs toward GAI chatbots. By grounding the selection of variables in qualitative findings, this study provides deeper insights into the key factors shaping GAI chatbot adoption. We have investigated context-specific enablers and inhibitors following prior studies (Park & Ryoo, 2013; Talwar et al., 2021). Furthermore, several studies also emphasize the significance of using different measures to determine trust and distrust (Cho, 2006; Lee et al., 2015). This study examines the impact of enablers and inhibitors on driving consumers' trust and distrust beliefs. By doing so, it extends DFT by providing a balanced perspective that differentiates the enabling factors that foster trust from the inhibiting factors that drive distrust. Finally, this study contributes to IT knowledge literature by confirming that IT knowledge moderates the relationships between consumer satisfaction and continued intention.

6.2. Practical implications

From a practical standpoint, this study provides valuable recommendations for e-commerce platforms aiming to improve consumer adoption and satisfaction with GAI chatbots. First, the positive impact of anthropomorphism and efficiency on trust

underscores the need for developing GAI chatbots that deliver personalized, natural interactions while resolving queries swiftly. Developers should continuously update algorithms to enhance chatbot performance through adaptive learning, ensuring chatbots meet diverse user needs effectively. Second, transparency plays a vital role in fostering trust. E-commerce platforms should clearly communicate data usage policies, provide real-time feedback on chatbot decisions, and define chatbot capabilities and limitations. Explainable AI features designed for GAI chatbots can help users understand how decisions are made, improving confidence in their reliability. Third, privacy concerns increase distrust, so e-commerce platforms need to implement proficient data protection protocols and explicitly explain these safeguards to consumers. For instance, the majority of Chinese e-commerce platforms have integrated blockchain, which offers privacy assurance facilities (Shahzad et al., 2024b). The marketing department should integrate such data protection technologies into GAI advertising to ensure consumers that their data is highly secure.

Fourth, emotional detachment can hinder trust, suggesting the need for chatbots to create more emotionally engaging interactions. Developers should use sentiment analysis and empathetic conversational styles to reduce impersonality, incorporating features like personalized greetings or adaptive tone modulation to foster a stronger connection with users. Fifth, enhancing IT knowledge can improve consumer interactions with GAI chatbots. Providing resources such as tutorials, guides, and FAQs can empower consumers to navigate chatbot features more effectively, bridging knowledge gaps and maximizing their benefits. Lastly, technical departments should actively monitor user feedback to address trust-related issues such as security concerns and performance inconsistencies. Regular improvements demonstrate a commitment to reliability, reducing distrust and encouraging long-term engagement with GAI chatbots.

6.3. Limitation and future work

This study provides significant insights into the factors influencing consumer trust and distrust in GAI chatbots. However, several limitations should be acknowledged. This study focuses on Chinese consumers, where cultural, regulatory, and online trust dynamics may differ among regions. Despite the findings providing valuable insights into GAI chatbot adoption in China, caution should be exercised when generalizing the results to other countries. Future studies could extend the research to diverse cultural contexts to enhance the external validity of the findings. The broader approach will help understand cultural differences and how they affect trust and distrust in GAI chatbots. The findings from China can also be validated in other regions by expanding the geographic scope. In addition, while the study conceptually distinguishes between trust and distrust as distinct constructs, the measurement items used reflect opposing statements. Although this approach aligns with prior research, it may not fully capture the subtle psychological differences between the two. Future research is encouraged to explore more nuanced measurement tools that can further clarify these conceptual boundaries. Furthermore, the current study is quantitative by design, so future studies can be conducted using mixed approaches to provide a more comprehensive understanding of GAI chatbots. Finally, future studies should consider additional enablers and inhibitors that may have been overlooked, such as perceived usefulness, ease of

use, and stress factors, to provide a more comprehensive understanding of their impact on consumer behavior.

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Compliance with ethical standards

This study was conducted according to the ethical standard of the institutional review board of universities/authorities and with the Helsinki declaration 1964, including updated recommendations.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Informed consent

Informed consent was attained from all individual respondents involved in the study.

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Appendix A. Questionnaire

Variables/items and source

Personalization (Lin et al., 2022)

- PR1: GAI chatbot provides suggestions and solves my queries as per my requirements.
 PR2: GAI chatbot predicts what kinds of products/ services I might want and makes suggestions
 PR3: GAI chatbot considers my unique preferences and circumstances.
 PR4: GAI chatbot delivers immediate one-on-one responses based on exactly what I demand.

Perceived anthropomorphism (Balakrishnan et al., 2022)

- PA1: GAI chatbot is natural; I do not feel fake about it.
 PA2: GAI chatbot is more humanlike.
 PA3: GAI chatbot is conscious of their actions.
 PA4: GAI chatbot feels lifelike and not artificial.
 PA5: GAI chatbot is elegant and engaging.

Efficiency (Hsu & Lin, 2023)

- EE1: GAI chatbot makes it easy to find the information I need.
 EE2: The information provided by GAI chatbot is well organized.
 EE3: GAI chatbot is simple to use.
 EE4: GAI chatbot provides well-organized information.
 EE5: GAI chatbot enables me to get information quickly.

Transparency (Shahzad et al., 2024a)

- TP1: The GAI chatbot provides clear and transparent information about how it processes user queries and makes decisions.
 TP2: The GAI chatbot takes responsibility for its responses and actions when assisting customers.
 TP3: The GAI chatbot provides detailed and understandable explanations about transactions and recommendations.
 TP4: I feel that I can provide feedback on my interactions with the GAI chatbot.

Privacy concerns (Adam & Benlian, 2024)

- PC1: I am sensitive about giving out information regarding my preferences.
 PC2: I am concerned about the anonymous information (information collected automatically but cannot be used to identify me, such as my computer, network information, operating system, etc.) that is collected about me.
 PC3: I am concerned about how my personally un-identifiable information (information that I have voluntarily given out but cannot be used to identify me, e.g. study subject, major, etc.) will be used by the firm.
 PC4: I am concerned about how my personally identifiable information (information that I have voluntarily given out AND can be used to identify me as an individual, e.g. email address, etc.) will be used by the firm.

Emotional detachment (Perrin-Martinénq, 2004)

- ED1: I feel frustrated with GAI chatbots' interactions.
 ED2: I feel irritated with GAI chatbots' interactions.
 ED3: I feel unfulfilled with GAI chatbots' interactions.
 ED4: I feel disconnected from GAI chatbots' interactions.
 ED5: I feel hostile towards GAI chatbots' interactions.

Trust (Johnson, 2007; Johnson et al., 2008)

- TT1: I can rely on GAI chatbots to execute my transactions reliably.
 TT2: Technology-related errors in GAI chatbots are quite rare.
 TT3: In my opinion, GAI chatbots are very reliable.

Distrust (Johnson, 2007; Johnson et al., 2008)

- DT1: I cannot rely on GAI chatbots to execute my transactions.
 DT2: Technology-related errors in GAI chatbots are quite frequent.
 DT3: In my opinion, GAI chatbots are not reliable.

IT knowledge (Shahzad et al., 2024b)

- ITN1: I have general knowledge of innovative technologies in e-commerce.
 ITN2: I have general knowledge of electronic data interchange.
 ITN4: I have general knowledge of cloud computing.
 ITN3: I have general knowledge of distributed ledgers.
 ITN5: I have general knowledge of the innovative system development life cycle.

Satisfaction (Bhattacharjee, 2001)

How do you feel about your overall experience of GAI chatbot use:

- SAT1: Very dissatisfied/Very satisfied.
 SAT2: Very displeased/Very pleased.
 SAT3: Very frustrated/Very contented.
 SAT4: Absolutely terrible/Absolutely delighted.

Continued intention (Bhattacharjee, 2001)

- CI1: I intend to continue using GAI chatbots rather than discontinue their use.
 CI2: I intend to continue using GAI chatbots than any alternative means
 CI3: I would like to continue my use of GAI chatbots.
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